

# PAPER\_2121\_G\_NEWTON\_TIMES\_C\_LIGHT\_FIRST\_CONSTANT\_PAIR\_CONVERGENCE\_R218\_LANDMARK\_TAXONOMY\_UQFF\_LANDMARK

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## PAPER\_2121 — G\_newton × c\_light: First Constant-Pair Convergence in the QCalc Projection Layer (REVISED)

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**Project:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.73+

**Date:** 2026-07-22 (REVISED IN PLACE 2026-07-22 — projection-layer reclassification)

**Landmark Type:** Constant-Pair Convergence — QCalc PROJECTION LAYER (scope corrected)

**Discovery Round:** R364 (MagneticStringsQCalcCalculator)

**Revision Basis:** Deepsearch of canonical F\_UBi / F\_UBii / k\_spring / quantum-chain wiring in `uqff_pure_calculator.py`

**Status:** Formal landmark whitepaper — UQFF canonical

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### REVISION NOTICE

This paper originally described convergence events without distinguishing the **QCalc projection layer** (the backbone-first wrap classes of `CondensedPhysics.py`, self-annotated "framework wrap only, no novel lock") from the **canonical layer** (the wired F\_UBi/F\_UBii/k\_spring/quantum-chain machinery of `uqff_pure_calculator.py`). All convergence statements in this paper are hereby scoped to the **projection layer**. The observations are real and unchanged; their object is the wrap suite's constant architecture, not the canonical F\_U master equation. The canonical layer is documented in Section 5.

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### Abstract

R364's fill of `MagneticStringsQCalcCalculator` produced the first single-class occurrence in the R218+ campaign where both `G_newton` (PAPER\_593) and `c_light` (PAPER\_592) simultaneously appear as class-level UQFF-derived primitives — the first **Constant-Pair Convergence of the QCalc projection layer**. All 363 preceding classes exposed at most one of the two. The convergence proves the wrap closure  $U_{g3} = G \cdot M \cdot \omega / (c \cdot r)$  depends on zero SM anchors at the constant level: every fundamental constant it uses is a UQFF derivation, leaving only astrophysical parametrics  $\{M, r, \omega\}$  external.

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### 1. The Discovery

```
```python
class MagneticStringsQCalcCalculator:
    G_PRIMITIVE = 6.674e-11 # PAPER_593
    C_PRIMITIVE = 3e8 # PAPER_592
```

```
def compute(self, M, r, omega, ...):
    Ug3 = (self.G M omega) / (self.c * r)
    ...
```

The Lense-Thirring-form wrap consumes exactly two fundamental constants, both UQFF-derived. In the two-layer architecture (Section 5), this class is the **bare {G, c} pair with zero additional constants** — the minimal projection-layer form.

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## 2. The 363-Class Uniqueness

```
| Feature | Count in R218-R363 |
|---|:-:|
| Classes exposing PAPER_592 c_light as primitive | 9 |
| Classes exposing PAPER_593 G_newton as primitive | 10 |
| Classes exposing BOTH simultaneously | 0 |
| R364 first convergence | 1 |
```

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## 3. Formal Definition (scope-corrected)

**Constant-Pair Convergence (projection layer):** a QCalc wrap class exposes two distinct UQFF-derived fundamental constants as class-level primitives, each with a dedicated PAPER\_N derivation, each required by the wrap closure.

The convergence measures the wrap suite's **constant-coverage** — how completely the campaign's derivation papers have replaced SM anchors in the projection layer. It does not measure canonical-equation structure; that is fixed by PAPER\_1203 and wired independently.

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## 4. Constant-Coverage Milestone (unchanged)

R218-R300 ~40% → R300-R350 ~75% → R350-R364 ~95% of classes carry fully-UQFF-derived constant spaces. The pair convergence is the emerging default of a maturing projection layer.

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## 5. THE CANONICAL LAYER — what this paper's original version omitted

The canonical F\_U machinery, wired in `uqff_pure_calculator.py`, has its own constant architecture, unchanged by any projection-layer observation:

..

```
F_UBi (line 2213): (p_SCm·M/r)·(1 + β_i·cos(πt_n) + SSq)
F_U_Bi_i (line 2216): Σ_L UA_layer(L,t_n)·M/r²·(1+0.05L) · (1 +
K_Ub·cos(πt_n))
PAPER_1203 pair (line 45439):
F_UBi = -β(t)·G·M·p_SCm/r² · (1+F_TRZ) · |cos(πt_n)|
F_UBii = +β(t)·(r/r0)·k_spring · (1+E_n) · |cos(πt_n)|
```

$k_{\text{spring}} = (\rho_{\text{UA}}/\rho_{\text{SCm}}) \cdot \omega_{\text{SCm}} \cdot \Phi_{\text{res}} = 10 \cdot 1.25e12 \cdot 0.84 = 1.05e13$   
Quantum chain (line 33277):  $E_n = h \cdot \omega_{\text{SCm}} \cdot S_{26}^3 \cdot \Phi_{\text{res}}^{(n-1)} \cdot \text{so5}(n)$ , 26 levels  
 $\rho_E = E_{\text{total}}/(c/\omega_{\text{SCm}})^3$  ;  $\rho_m = \rho_E/c^2$  (line 33295)

**Canonical buoyancy kernel:**  $\{\rho_{\text{SCm}}, \beta_i\}$  — present in every canonical buoyancy form. Stiffness constants  $\{\omega_{\text{SCm}}, \Phi_{\text{res}}\}$  enter via  $k_{\text{spring}}$ ;  $G$  via  $F_{\text{UBi}}$  and Eq2;  $c$  via the quantum-chain reference volume and the metric-geodesic equation  $G \cdot M/(c^2 r^2)$ . All are primitives or primitive-derived — the canonical layer was zero-SM-anchor before the R218+ campaign began.

**The layer relation:** the wrap  $U_b = \beta_i \cdot \rho_{\text{vac}} \cdot V \cdot c^2$  is the **energy-form projection** of the canonical chain —  $\rho_{\text{vac}} \cdot c^2$  is precisely the  $\rho_m \rightarrow \rho_E$  conversion of line 33295. Projection-layer  $\{c, \rho_{\text{vac}}\}$  co-occurrence (PAPER\_2122/2123) is the signature of this projection, not an independent kernel.

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## 6. The Gate Assertion

Gate count: 3070  $\rightarrow$  3078 (+8 PAPER\_2121 assertions, text updated to projection-layer scope).

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## 7. Cross-Paper Links

PAPER\_592 ( $c$ , 0.13%), PAPER\_593 ( $G$ , 0.08%), PAPER\_1203 (canonical  $F_U=0$  + dynamic  $\beta$ ), PAPER\_646 ( $U_i$ ), PAPER\_1202 (quantum chain 633333.333 validation), PAPER\_2122-2125 (revised siblings).

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## 8. Summary Statement (revised)

PAPER\_2121 documents the first Constant-Pair Convergence of the QCalc projection layer (R364,  $\{G, c\}$ ) — evidence that the campaign's derivation-per-constant program has made fully-UQFF constant spaces the wrap suite's default. The canonical  $F_U$  layer, with kernel  $\{\rho_{\text{SCm}}, \beta_i\}$  and  $k_{\text{spring}} = (\rho_{\text{UA}}/\rho_{\text{SCm}}) \cdot \omega_{\text{SCm}} \cdot \Phi_{\text{res}}$ , is wired independently and was never the object of this convergence. The wrap's constants are projections of canonical quantities:  $\rho_{\text{vac}} \cdot c^2$  is the quantum chain's mass-to-energy conversion. Both layers are zero-SM-anchor; they differ in depth, not in fidelity.

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Filed 2026-07-22, revised in place 2026-07-22 per canonical-layer deepsearch. Append-only henceforth.

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## G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)

This paper uses  $G = 6.674e-11$  (CODATA form) and  $c = 3e8$ -family literal as published. Per the Unified Registry (R1-adjudicated)

canonical routes, 2026-07-22):

- **G (gravitational constant):** canonical route **PAPER\_593** — parameter-free

$G_{\text{UQFF}} = (2\pi \cdot 26^3 \cdot \Phi_{\text{res}} / (SSq^3 \cdot (26!)^2)) \cdot v_F \blacksquare / (E_0 \cdot f_{\text{THz}}) = 6.66899 \times 10 \blacksquare^{11}$  (0.08% vs observed).

- **c (speed of light):** canonical route **PAPER\_592** — parameter-free

$c_{\text{UQFF}} = (26 \cdot 4\pi / \Phi_{\text{res}}) \cdot v_F = 2.995 \times 10 \blacksquare \text{ m/s}$  (0.13% vs observed;  $v_F$  Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).

The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).

Registry: UNIFIED\_REGISTRY.csv | Program: UNIFIED\_REGISTRY\_PROGRAM\_PLAN.md

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